Muhammed Shanshad P K

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.32
S.P.B.H.S.S RAMANATTUKARA	Class XII (GOVERNMENT OF KERALA BOARD OF HIGHER SECONDARY EXAMINATION)	2017	88.75%
S.P.B.H.S.S RAMANATTUKARA	Class X (BOARD OF PUBLIC EXAMINATIONS, KERALA)	2015	97.00%

EXPERIENCE

CoFounder - InSight | 2023 -24

- **Managed end-to-end client relationships** and onboarding for entrepreneurs applying to loan-linked subsidy schemes
- **Translated business requirements into structured financial documentation** including comprehensive project reports and projected balance sheets tailored to subsidy scheme criteria.
- Delivered **48+ financial projects worth INR 42.3 million** in total, achieving a **95.8% approval rate** across client engagements.

Entrepreneurship Development Executive | Department of Industries & Commerce, Government of Kerala | 2022 - 23

- **Led entrepreneurship development activities for Peruvayal Gram Panchayat**, guiding aspiring and existing entrepreneurs on government finance schemes, licensing, subsidies, and business-support programs, including facilitating applications.
- **Organized Loan, License & Subsidy Fests** and conducted field-level outreach with aspiring entrepreneurs, providing one-to-one guidance, facilitating applications, and improving access to government schemes and institutional support.
- **Collaborated with banks, financial institutions, government departments, training institutions, and incubation centres** including RSETI, NIT incubation initiatives, and ICAR-IISR to connect entrepreneurs with financing, training, incubation, and technical support.
- **Supported the establishment of an Entrepreneurship Development (ED) Club at AWH Engineering College**, promoting entrepreneurship among students through awareness and institutional engagement.
- **Facilitated INR 4.82 million in subsidy-linked financing** for entrepreneurs by guiding scheme applications and coordinating with banks, financial institutions, and government authorities.

PROJECTS

Autonomous Mapping Robot (ROS 2) | Github Link

- Built autonomous frontier-based exploration and mapping for a 4-wheel differential-drive robot in **ROS 2 Jazzy** and **Gazebo Harmonic**, combining **SLAM Toolbox** for mapping, **Nav2** for navigation and **explore_lite** for frontier selection, so the robot maps unknown environments with no manual driving.

4-Wheel Differential Drive Robot | *Github Link*

- **Built and mapped a 4-wheel differential-drive robot** using DDSM210 motors and Raspberry Pi 5, integrating ROS 2 Jazzy for motor control, sensor integration, and SLAM-based mapping during practical testing and operation.

Velocity-Controlled Quadruped Walking | Github Link

- Trained an 8-DOF quadruped with closed-loop five-bar-linkage legs to walk under velocity commands using **Reinforcement Learning**, training a PPO policy with Stable-Baselines3 in MuJoCo with a trot gait clock, domain randomization, and keyboard control.

Sensor Fusion & SLAM | *Github Link*

- **Developed a drift-reduced SLAM pipeline** by fusing IMU and LiDAR data using the ROS 2 Robot Localization package, improving localization consistency and maintaining reliable map generation during robot navigation

RL-Based Two-Wheel Robot Stabilization | *Github Link*

- **Stabilized a two-wheel robot in simulation using Reinforcement Learning**, training a PPO policy with Stable-Baselines3 in MuJoCo to learn dynamic balance through continuous interaction with the simulated environment in practice.

Unmanned Surface Vehicle (USV)

- **Designed and fabricated a 64-kg-capacity unmanned surface vehicle** using 3D-printed components, integrating Pixhawk X7 Pro and ArduPilot Mission Planner for vehicle control and autonomous mission planning during field testing.

2-DOF Robot Manipulator Control | *Github Link*

- **Derived the mathematical model and designed a control system from first principles** for a 2-DOF robotic manipulator, implementing the controller and validating its performance through MATLAB/Simulink simulation runs.

THESIS

Agile Locomotion of a 12-DOF Quadruped Using Reinforcement Learning

- **Designed the RL-based locomotion methodology** for a 12-DOF quadruped, targeting agile and adaptive locomotion through reinforcement learning.
- **Developing and training an 8-DOF quadruped prototype in MuJoCo**, using **PPO with Stable-Baselines3** to learn locomotion policies through simulated interaction.
- **Developing the simulation-to-real transfer pipeline**, with the trained locomotion policy intended for deployment on the physical quadruped prototype.

RELEVANT COURSES

Robotics : Autonomous Robotic Systems, Marine Robotics, Soft Robotics, Control Systems Design

AI & Computational Methods: Machine Learning, Reinforcement Learning, Probability and Optimization

Laboratory: Sensors & Actuators Lab, Robotics & Automation Lab, Computer Vision Lab, Advanced Robotics Lab

SKILLS

Programming Languages: Python, SQL, C++

Tools & Libraries : ROS2 Humble, ROS2 Jazzy, MATLAB, Gazebo Classic, Gazebo Harmonic, MuJoCo, URDF, Pandas, Numpy, Scikit-learn, OpenCV, Fusion 360, Arduino, Stable-Baselines3,TensorFlow,PyTorch

CERTIFICATIONS

ROS 2 From Scratch -- Packt, Coursera

Post Graduate Certificate Program in Entrepreneurship: Small Business Management (PGCSBM) -- Digital University of Kerala

POSITIONS OF RESPONSIBILITY

Core Member & Mentor: Robotix Club

Student Coordinator: Placement Cell

- Engaged with companies and recruitment teams to facilitate placement and internship opportunities.
- Coordinated student-Placement Cell communication and disseminated recruitment updates, eligibility criteria, and deadlines.
- Ensured timely access to relevant opportunities and information for the student batch.

Placement Training Volunteer

- Prepared mock placement examination question papers for the 2026 batch to support structured placement preparation

Technical Coordinator & Mentor - Autonomous Robotics Summer Immersion & Simulation Grand Prix 2026

- Coordinated a **10-day residential robotics workshop** by **IIT Hyderabad and George Mason University**.
- Taught **ROS 2 Jazzy, RViz2, and Gazebo Harmonic** through hands-on simulation sessions.
- **Mentored participants and troubleshot** simulation-based robotics tasks.

EXTRACURRICULAR

Silver Medalist - Kickboxing (Category 2), DIESTA 2026, representing the **MAE Department**.

Gold Medalist, Football, Milan 2026, as Starting XI player representing **Bhabha Hostel**.

HOBBIES

Football, Guitar, Reading, Swimming, MMA